



Cal Poly Pomona

ECE 3300 Lecture

Pac-man Game

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Description of the System

For this project, our team designed and implemented a Pac-Man style game using Verilog on the Nexys A7 FPGA board. The purpose of the project was to create a complete FPGA-based digital system that uses the board's inputs and outputs in a meaningful way. The game displays a maze, Pac-Man, moving objects, score information, and background effects through the VGA output. The player controls Pac-Man using the push buttons, while the switches are used to control game settings such as speed.

The system is organized using a top-level module that connects several smaller modules together. Each module has a specific job, which makes the design easier to understand and debug. The main parts of the system include the VGA timing generator, map controller, movement logic, collision detection, score controller, LED output logic, seven-segment score display, UART transmitter, and FIFO. The top-level module connects these parts so they work together as one complete game system.

The map contains wall tiles, open spaces, and dot locations. The collision detection logic checks Pac-Man's current position against the map so Pac-Man cannot move through walls. The score controller updates the score when dots are collected and also controls game conditions such as lives remaining, power mode, game over, and win status. This makes the game more than just a simple display because the logic responds to player movement and game conditions.

The system also includes UART communication. The UART portion of the project converts score information into ASCII characters, stores those characters in a FIFO, and then transmits them through the UART transmit pin. This allows the FPGA to send game status information to a serial terminal. Overall, the project is a complete digital system because it includes real board inputs, real board outputs, multiple Verilog modules, VGA display logic, UART communication, FIFO logic, FSM logic, and a master testbench for verification.

Explanation of I/O's

The switches are used as real control inputs for the Pac-Man game. The lower switches control the game speed, which allows the player to change how fast Pac-Man moves during gameplay. These switch inputs are meaningful because they directly affect the behavior or appearance of the game instead of simply being connected without a purpose. During the hardware demo, the switch operation can be shown by changing the speed setting and observing the corresponding LED output and gameplay behavior.

The push buttons are used to control Pac-Man's movement and to reset the game. The directional push buttons control movement up, down, left, and right, while the reset button restarts the system. These buttons are part of the actual game behavior because the player uses them to move Pac-Man through the maze. The testbench also verifies that each push button produces the correct movement direction code, which helps confirm that the button logic is working correctly.

The LEDs are used to display game status information. Some LEDs show the selected speed setting, remaining lives, and win status. This makes the LEDs useful during gameplay because they provide quick visual feedback about the current state of the game. The RGB LED is also used as a game-state indicator. In normal gameplay, the RGB LED shows green, and when the game is over, it shows red. This gives the player a simple way to identify the game state without only relying on the VGA screen.

The seven-segment display is used to show the player's score. The score is separated into thousands, hundreds, tens, and ones digits. The display uses multiplexing so that the digits appear active at the same time, even though the FPGA is actually scanning through the digits quickly. This is important because the display shows real game information instead of a random or unused value.

The UART is used to transmit score and status information from the FPGA. In this design, the UART is transmit-only because the game inputs are handled by the push buttons and switches. The UART transmit logic takes the score digits, converts them into ASCII characters, stores them in a FIFO, and sends them out through the UART transmit signal. This allows the project to demonstrate serial communication by sending game information to an external serial terminal.

The VGA output is the main display for the game. It shows the maze, background effects, Pac-Man, moving objects, and score. The VGA timing module generates the horizontal sync, vertical sync, and pixel position signals needed to draw the game on a monitor. The display uses multiple layers, and the final pixel color is selected based on priority. For example, the score/message layer has priority over the moving objects, the moving objects have priority over the maze, and the maze has priority over the background. This allows the game screen to display multiple objects clearly.

FSM

The project uses finite state machines, especially in the UART section of the design. A finite state machine is a circuit that moves through a set of defined states based on the clock and input conditions. This is useful when a process must happen in a specific sequence. In this

project, the UART transmitter uses an FSM because serial communication must send data in a strict order.

The UART transmitter FSM waits in an idle state until there is data ready to send. Once transmission begins, the FSM sends a start bit, then sends the eight data bits one at a time, then sends a stop bit, and finally returns to the idle state. This sequence is necessary because UART communication requires the receiving device to recognize the start bit, sample each data bit, and then confirm the stop bit. Without the FSM, the UART transmitter would not know which part of the byte it is currently sending.

The UART status logic also uses state-based control to decide when characters should be loaded into the FIFO and when they should be sent to the UART transmitter. This allows the message-loading logic and the UART transmission logic to work together without sending characters out of order. The FSM is important because it controls timing and sequence, which are both required for reliable UART transmission.

FIFO

The project uses a FIFO as part of the UART system. FIFO stands for First In, First Out, which means the first data value written into the FIFO is the first value read out. In this project, the FIFO stores ASCII characters before they are transmitted by UART. For example, when the score is converted into ASCII characters, those characters are written into the FIFO in the correct order. The UART transmitter then reads the characters from the FIFO one at a time and sends them through the UART transmit line.

The FIFO is useful because UART transmission is much slower than the FPGA clock. The FPGA can prepare and load characters into the FIFO, while the UART transmitter sends them out at its slower baud rate. This prevents the message-loading logic from needing to wait directly on every UART bit. It also helps keep the design organized because one part of the circuit prepares the data while another part transmits it.

The testbench verifies the FIFO by writing values into it and then reading them back out. The simulation checks that the FIFO is empty after reset, becomes full after 16 writes, and becomes empty again after 16 reads. It also verifies that the data comes out in the same order it was written. This confirms that the FIFO is behaving correctly as a First In, First Out storage system.

Modeling Styles Used

This project uses four Verilog modeling styles: gate-level modeling, structural modeling, behavioral modeling, and dataflow modeling. Gate-level modeling is used in the LED output

logic with basic logic gate primitives such as buffers, OR gates, and NOT gates. This shows that part of the design is described using low-level gate connections.

Structural modeling is used throughout the project because the top-level module connects many smaller modules together. The top-level design instantiates the clock manager, VGA timing module, background effects module, map controller, moving object logic, score controller, LED output module, seven-segment display module, and UART status transmitter. This is structural modeling because the complete system is built by connecting submodules together through their ports.

Behavioral modeling is used with always blocks to describe how the circuit behaves over time. The project uses sequential always @(posedge clk) blocks for clocked logic and combinational always @(*) blocks for logic that responds immediately to input changes. Behavioral modeling is used in areas such as movement control, score updating, FSM operation, and display-selection logic.

Dataflow modeling is used with continuous assign statements. These statements describe direct relationships between signals. For example, dataflow modeling is used to assign VGA color outputs, FIFO full and empty flags, and other simple signal connections. This style is useful for combinational logic that can be described with direct signal assignments.

Verilog Language Features

The design uses ``define` to create a named constant in the LED logic, which makes the code easier to read and change. The design also uses parameter in reusable modules such as the FIFO and UART transmitter. Parameters make the modules more flexible because values such as data width, address width, or UART clock timing can be changed without rewriting the full module.

The project uses `localparam` for internal constants such as VGA colors, screen offsets, and FSM state values. These constants are used inside specific modules and help make the code more readable. The project also uses a for-generate loop in the LED output module to create repeated hardware connections for multiple LED bits. This avoids writing the same type of gate connection multiple times.

The testbench uses for loops to run repeated test cases. One example is the switch and LED test, where the testbench checks every speed setting from 0000 to 1111. This helps verify that all switch combinations are handled correctly instead of only testing one or two cases.

Testbench

The project uses a master testbench that instantiates the full top-level design. This means the simulation tests the complete system rather than only testing one small module by itself. The purpose of the testbench is to verify that the design behaves correctly before running it on the FPGA board. The testbench checks both activity and correctness. Activity checks confirm that signals are changing, while correctness checks confirm that outputs match expected behavior.

The testbench uses reusable tasks to make the simulation more organized. Tasks are used for actions such as applying reset, pressing buttons, checking speed LEDs, receiving UART data, and printing system status. Using tasks makes the testbench easier to read because repeated actions do not need to be written many times. It also makes the verification process more organized and easier to explain.

The simulation verifies several important parts of the system. It checks that reset correctly restores lives, game-over status, and win status. It checks that the switches correctly control the LED speed output. It verifies that the map ROM contains wall tiles, dot tiles, and open spaces. It also checks that the score increases after Pac-Man collects a dot. These tests prove that the game logic is responding correctly to expected conditions.

The testbench also verifies the push-button direction logic. Each button is pressed in simulation, and the testbench checks that the correct movement direction code is produced. This confirms that the button inputs are connected to the movement system correctly. In addition, the testbench includes a wall-collision check. It places Pac-Man on an open tile and confirms that moving left is blocked by a wall while moving right is allowed. This shows that the collision logic is working and that Pac-Man is not simply allowed to move through every tile.

The FIFO is tested by writing data into it and reading the data back out. The testbench checks that the FIFO is empty after reset, full after 16 writes, and empty again after 16 reads. It also checks that the values come out in the same order they were written. This proves that the FIFO follows First In, First Out behavior.

The UART testing is split into two parts. First, the testbench verifies that score digits are correctly converted into ASCII characters. Then it checks that the full design produces UART transmit activity and that the UART line returns to the idle-high state. This confirms that the UART system is active in the complete design. The VGA test checks that horizontal sync and vertical sync are active and that the map layer, moving object layer, score/message layer, and final VGA color outputs produce visible activity.

At the end of the simulation, the testbench prints a final summary of the major signals and test results. The most important result is that the simulation ended with `error_count = 0` and printed **OVERALL RESULT: PASS** - no testbench errors detected. This means the complete design passed the verification checks included in the master testbench.

```
score=0010 lives=2 won=0 over=0 power=0 leds=01100000 rgb=R0G1B0
PASS: lives reset to 2 remaining.
PASS: game_won reset correctly.
PASS: game_over reset correctly.
TEST 2: Speed switch/LED sweep using for loop
PASS: speed 0000 shown on LEDs.
PASS: speed 0001 shown on LEDs.
PASS: speed 0010 shown on LEDs.
PASS: speed 0011 shown on LEDs.
PASS: speed 0100 shown on LEDs.
PASS: speed 0101 shown on LEDs.
PASS: speed 0110 shown on LEDs.
PASS: speed 0111 shown on LEDs.
PASS: speed 1000 shown on LEDs.
PASS: speed 1001 shown on LEDs.
PASS: speed 1010 shown on LEDs.
PASS: speed 1011 shown on LEDs.
PASS: speed 1100 shown on LEDs.
PASS: speed 1101 shown on LEDs.
PASS: speed 1110 shown on LEDs.
PASS: speed 1111 shown on LEDs.
```

Figure 1: Reset, speed switch, and LED verification.

Figure 1 shows the beginning of the simulation results. The testbench first verifies that the game resets correctly by checking that the lives reset to two, `game_won` resets to 0, and `game_over` resets to 0. The simulation then checks the speed switch inputs using a for loop. Each switch value from 0000 to 1111 is tested, and the output shows that each speed value is correctly displayed on the LEDs. This verifies both the reset logic and the switch-to-LED behavior.

```

TEST 4: Map/wall/dot verification
Map count: walls=548 open=320 dots=258 first_dot=(col=1,row=1)
PASS: map_rom reports wall tiles.
PASS: map_rom reports dot tiles.
PASS: no-dot region row 10 col 8 is not a dot.
PASS: row 0 col 0 is a wall/border tile.
TEST 5: Score change from dot collection
Score before=10 after=20
PASS: score increased after dot collection.
TEST 6: Push-button direction correctness
PASS: UP button correctly gives move_dir=0100
PASS: LEFT button correctly gives move_dir=0001
PASS: RIGHT button correctly gives move_dir=0010
PASS: DOWN button correctly gives move_dir=1000
Movement check: before=(208,368) after=(208,368)
WARNING: Pac-Man position did not change. Direction logic passed, but movement may be blocked by a wall or need longer simulation.
TEST 6B: Wall-collision correctness
PASS: Pac-Man placed at open tile (1,1).
PASS: LEFT is correctly blocked by the wall at tile (0,1). valid_moves=1010
Wall test: before=(16,16) after=(16,16)
PASS: Pac-Man stayed in place when commanded into a wall.
PASS: RIGHT is correctly open from tile (1,1). valid_moves=1010
PASS: Wall-collision test complete: LEFT blocked, RIGHT open.

```

Figure 2: Map ROM, score, push-button direction, and wall-collision verification.

Figure 2 shows the simulation tests for the main Pac-Man game logic. The testbench checks that the map contains wall tiles, open spaces, and dot tiles. It also confirms that the border tile is detected as a wall. The score test shows that the score increased from 10 to 20 after dot collection, which verifies that the score controller responds correctly when Pac-Man collects a dot. The push-button test verifies that each button produces the correct movement direction code for up, left, right, and down. The wall-collision test confirms that Pac-Man is blocked when moving toward a wall and that an open direction is still allowed. This proves that the movement and collision logic are working correctly.

```

TEST 7: Seven-segment activity
PASS: seven-segment anodes scanned. an_edges=2006
PASS: seven-segment segment outputs changed. seg_edges=1004
TEST 8: Power mode LED
PASS: LED4 turns on during power mode.
TEST 9: Lives LEDs and RGB game-over status
PASS: 2 lives shown correctly.
PASS: 1 life shown correctly.
PASS: 0 lives shown correctly.
PASS: RGB green during play.
PASS: RGB red during game over.
TEST 10: Winner state
PASS: LED7 turns on during WINNER.

```

Figure 3: Seven-segment display, power mode LED, lives LEDs, RGB LED, and winner-state verification.

Figure 3 shows the simulation results for several board output devices. The seven-segment display test confirms that the anodes are scanning and that the segment outputs are changing, which shows that the score display is being multiplexed correctly. The power mode test confirms that LED4 turns on during power mode. The lives LED test verifies that the LEDs correctly show two lives, one life, and zero lives. The RGB LED test confirms that the RGB LED is green during normal gameplay and red during game over. The winner-state test confirms that LED7 turns on when the game reaches the win condition.

```
TEST 11: FIFO full/empty and data order behavior
PASS: FIFO empty after reset.
PASS: FIFO count is 0 after reset.
FIFO count after writes = 16
PASS: FIFO full after 16 writes.
PASS: FIFO count is 16 after 16 writes.
PASS: FIFO read index 0 correct. data=41
PASS: FIFO read index 1 correct. data=42
PASS: FIFO read index 2 correct. data=43
PASS: FIFO read index 3 correct. data=44
PASS: FIFO read index 4 correct. data=45
PASS: FIFO read index 5 correct. data=46
PASS: FIFO read index 6 correct. data=47
PASS: FIFO read index 7 correct. data=48
PASS: FIFO read index 8 correct. data=49
PASS: FIFO read index 9 correct. data=4a
PASS: FIFO read index 10 correct. data=4b
PASS: FIFO read index 11 correct. data=4c
PASS: FIFO read index 12 correct. data=4d
PASS: FIFO read index 13 correct. data=4e
PASS: FIFO read index 14 correct. data=4f
PASS: FIFO read index 15 correct. data=50
FIFO count after reads = 0
PASS: FIFO empty after 16 reads.
PASS: FIFO count returned to 0 after reads.
```

Figure 4: FIFO full, empty, count, and data-order verification.

Figure 4 shows the FIFO test results. The testbench first confirms that the FIFO is empty after reset and that the FIFO count is 0. It then writes 16 values into the FIFO and confirms that the FIFO becomes full and that the count reaches 16. After that, the testbench reads the values back out and checks each read value. The data comes out in the correct order from 41 through 50 in hexadecimal, which corresponds to ASCII characters A through P. This confirms that the FIFO is working correctly as a First In, First Out memory.

```

TEST 13: Full-design UART TX activity
PASS: UART TX toggled in full design. uart_edges=28
PASS: UART line returned to idle-high.
TEST 14: VGA and display-layer activity
frame=1 score=0010 lives=2 won=0 over=0 power=0 speed=1111 leds=01101111 an=11101111 seg=1111111
PASS: hsync toggled. hsync_edges=2029
PASS: map layer drew non-black pixels. count=471120
PASS: moving object layer drew non-black pixels. count=2724
PASS: score/message layer drew non-black pixels. count=7192
PASS: VGA color outputs became nonzero. count=445466
PASS: vsync toggled. vsync_edges=2
TEST 15: Final reset
score=0010 lives=2 won=0 over=0 power=0 leds=01101111 rgb=R0G1B0

```

Figure 5: UART and VGA simulation verification.

Figure 5 shows the UART and VGA tests. The UART test confirms that the UART transmit signal toggles in the full design and then returns to the idle-high state. This proves that the UART transmitter is active in the complete system. The VGA test confirms that the horizontal sync signal toggles, the vertical sync signal toggles, and the VGA display layers produce visible pixel activity. The map layer, moving object layer, score/message layer, and final VGA color outputs all produce nonzero pixel counts. This shows that the VGA system is drawing the game display during simulation.

```

TEST 14: VGA and display-layer activity
frame=1 score=0010 lives=2 won=0 over=0 power=0 speed=1111 leds=01101111 an=11101111 seg=1111111
PASS: hsync toggled. hsync_edges=2029
PASS: map layer drew non-black pixels. count=471120
PASS: moving object layer drew non-black pixels. count=2724
PASS: score/message layer drew non-black pixels. count=7192
PASS: VGA color outputs became nonzero. count=445466
PASS: vsync toggled. vsync_edges=2
TEST 15: Final reset
score=0010 lives=2 won=0 over=0 power=0 leds=01101111 rgb=R0G1B0

```

Figure 6: Output waveform for VGA, LEDs, RGB LED, and seven-segment display

Figure 6 shows the simulated waveform outputs for the main board signals. The VGA red, green, and blue signals change over time, which shows that the display logic is producing pixel color data. The VGA horizontal sync and vertical sync signals are present, which confirms that the VGA timing module is active. The LED output shows the current game status, including the selected speed and game-state information. The RGB LED output shows green during normal gameplay. The seven-segment segment and anode signals also change, which shows that the score display is being scanned correctly.

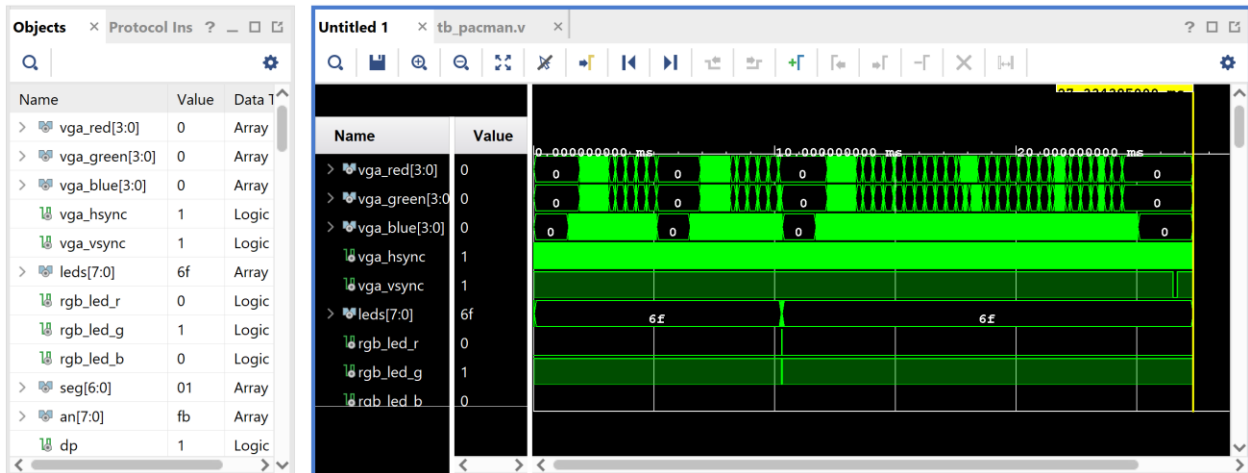


Figure 7: Simulation waveform showing VGA color outputs, VGA sync signals, LEDs, RGB LED, seven-segment segments, anode scanning, and decimal point activity during the Pac-Man testbench

Figure 7 shows the simulated output waveforms for the main board outputs. The VGA red, green, and blue signals change over time, which shows that the display logic is producing pixel color data. The VGA horizontal sync and vertical sync signals are also present, which confirms that the VGA timing module is active. The LED output shows the current game status, including speed and game-state information. The RGB LED output shows green during normal gameplay. The seven-segment segment and anode signals also change, which shows that the score display is being multiplexed correctly.

```
Tcl Console x Messages Log
[Icons]
TEST 15: Final reset
score=0010 lives=2 won=0 over=0 power=0 leds=01101111 rgb=R0G1B0
-----
Final Testbench Summary
error_count      = 0
hsync_edges      = 2030
vsync_edges      = 2
uart_edges       = 29
an_edges         = 5339
seg_edges        = 2671
map_pixel_count  = 471120
moving_pixel_count = 2724
score_pixel_count = 7192
vga_color_count  = 445466
dot_count        = 258
wall_count       = 548
open_count       = 320
leds             = 01101111
seg              = 0000001
an               = 11111011
rgb              = R0 G1 B0
score=0010 lives=2 won=0 over=0 power=0 leds=01101111 rgb=R0G1B0
-----
OVERALL RESULT: PASS - no testbench errors detected.
$stop called at time : 27334385 ns : File "C:/Users/castr/Desktop/ECE 3300/Pacman Game/pacman_nexys_a7_v3/src/tb_pacman.v" Line 1055
run: Time (s): cpu = 00:00:26 ; elapsed = 00:02:21 . Memory (MB): peak = 3576.641 ; gain = 1657.238
```

Figure 8: Final simulation summary.

Figure 8 shows the final testbench summary. The simulation reports `error_count = 0`, which means none of the testbench checks failed. The summary also shows the final counts for VGA sync activity, UART activity, seven-segment activity, map pixels, moving-object pixels, score pixels, and map tile counts. The final line states `OVERALL RESULT: PASS - no testbench errors detected`. This confirms that the complete design passed the simulation checks.

In conclusion, this project successfully implemented a Pac-Man style game on the Nexys A7 FPGA board using Verilog. The final design used the board I/Os, including switches for speed control, push buttons for movement and reset, LEDs and RGB LEDs for game status, the seven-segment display for the score, UART for transmitting score/status information, and VGA for displaying the game. The project also included Verilog concepts such as FSM logic, FIFO storage, gate-level modeling, structural modeling, behavioral modeling, dataflow modeling, parameters, localparams, a ``define` statement, and a `for-generate` loop. The final testbench verified the main parts of the design, including reset behavior, button direction control, wall collision, scoring, FIFO order, UART activity, LED outputs, seven-segment display activity, and VGA output. The simulation passed with zero errors, which showed that the design worked correctly in simulation.

This project helped us better understand how a larger FPGA system is built by connecting smaller modules together. We learned how important it is to keep each module organized so the top-level design is easier to follow and debug. One challenge was making sure the

simulation checked actual correctness instead of only checking that signals were changing. Another issue was working through timing differences in the testbench, especially for movement and UART, because some signals do not update immediately in simulation. After fixing those testbench issues, the final simulation results matched the expected behavior. Overall, this project showed how Verilog can be used to create a complete digital system with game logic, display output, communication, and verification.